Industrial 2-Finger Parallel Gripper

CAD Design, Load Calculations and Static Stress Planning

Item	Detail
Student	Vedant Deshmukh
Programme	ROBO AI Industrial Training Program on Robotics and AI (My Equation) – Mini Project
Mechanism	Rack-and-pinion parallel-jaw gripper, single central pinion
CAD software	SolidWorks Design Premium 2026 SP0.0
Payload range	5 kg to 10 kg
Colour scheme	Housing and pinion: yellow. Fingers: blue.
Report version	v1.0 – November 2026

1. Project Status

This report documents version 1 of the gripper. The CAD model, assembly and load calculations are complete. The ANSYS static structural simulation has **not yet been performed**; this report therefore gives hand-calculated stress checks only, together with the prepared simulation setup. Known issues in the v1 model are stated openly in Section 8 along with the planned fixes.

Requirement from problem statement	Status
Parametric CAD of fingers, base and actuation interface	Done (SolidWorks)
Payload capability 5–10 kg	Grip force and pinion torque calculated
Assembly with motion constraints and jaw symmetry	Assembled with mates; second finger orientation to be corrected
Yellow and blue colour scheme	Done
Static structural simulation (ANSYS)	**Pending.** Hand-check done, FEA setup prepared
Safety factor of at least 1.5	Applied to grip force; stress factor of safety to be confirmed by FEA

2. Conceptual Design (Question 1)

The gripper is a parallel-jaw gripper driven by one central pinion that meshes with two racks on opposite sides of it. Each rack is rigidly attached to one finger. When the pinion turns, its two contact points move in opposite directions, so one rack moves one way and the other moves the opposite way by the same distance. The two fingers therefore open and close symmetrically about the pinion axis. The walls and floor of the U-shaped housing channel guide each finger along a straight line, which keeps the finger plates parallel.

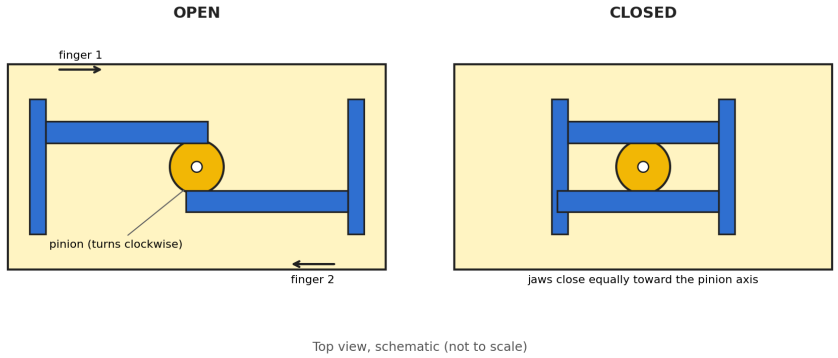


Figure 1: Concept diagram – one pinion drives two racks, so both jaws move equally and oppositely.

Rack travel per pinion revolution equals the pitch circumference: π × m × z = π × 1.25 × 16 ≈ 62.8 mm per finger.

3. Gear Design Basis

Parameter	Value	Note
Module (m)	1.25 mm	Same for pinion and racks
Pinion teeth (z)	16	
Pressure angle (nominal)	20°	Not modelled; teeth are rectangular
Pitch diameter	20 mm	m × z
Outside diameter	22.5 mm	m × (z + 2)
Tooth pitch	3.927 mm	π × m
Rack tooth-tip line to pinion axis	8.75 mm	m × (z/2 − 1)
Tooth form modelled	Rectangular notches	Simplified, see Section 8

The teeth are modelled as rectangular notches, not involute profiles. This keeps the model quick to build and gives the correct pitch and meshing positions, but it is the main cause of the interference reported in Section 5.

4. CAD Sketching and 3D Modelling (Questions 2 and 3)

Part	Key features and dimensions	Colour
Housing_02	Base block 140 × 76 mm (Boss-Extrude). Pocket sketch 100 × 60 mm, then further cuts, fillet and chamfer to form the U-shaped channel. 4 mm centre hole for the pinion axis.	Yellow
FINGER_01 (used twice)	Plate 60 × 50 mm with neck, fillet and chamfer. Rack bar 60 mm long × 12.5 mm wide, tooth edge 8.75 mm from plate centreline. Rack notch 2.6 mm wide × 2.8 mm deep, linear pattern of 10 at 3.927 mm spacing.	Blue
PINION	Boss-Extrude: Ø22.5 mm disc, 8 mm thick (mid-plane). Notch 2 × 2.8 mm, circular pattern of 16, equal spacing. Ø4 mm centre bore.	Yellow

All parts are built from dimensioned sketches and parametric features (extrude, cut, fillet, chamfer, linear and circular patterns), so changing a sketch dimension updates the 3D part and the tooth patterns.

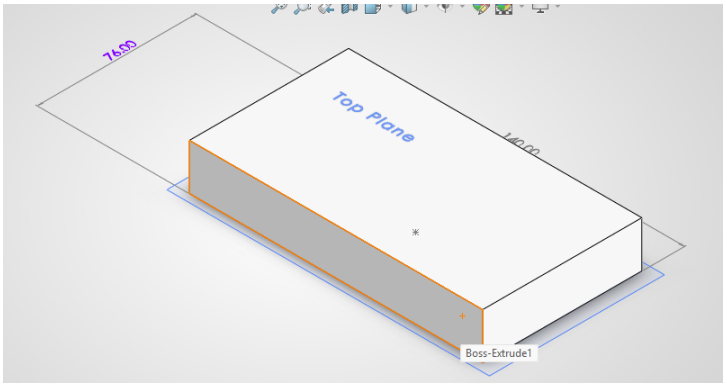


Figure 2: Housing – base block, 140 × 76 mm (Boss-Extrude1).

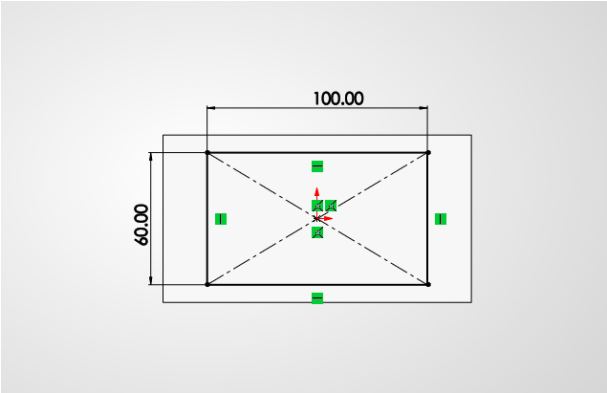


Figure 3: Housing pocket sketch, 100 × 60 mm, centred on the origin.

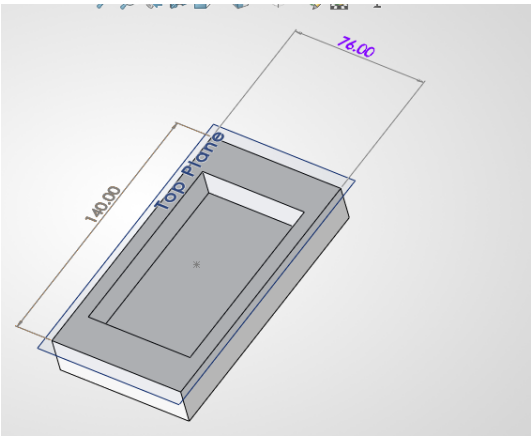


Figure 4: Housing after the pocket cut.

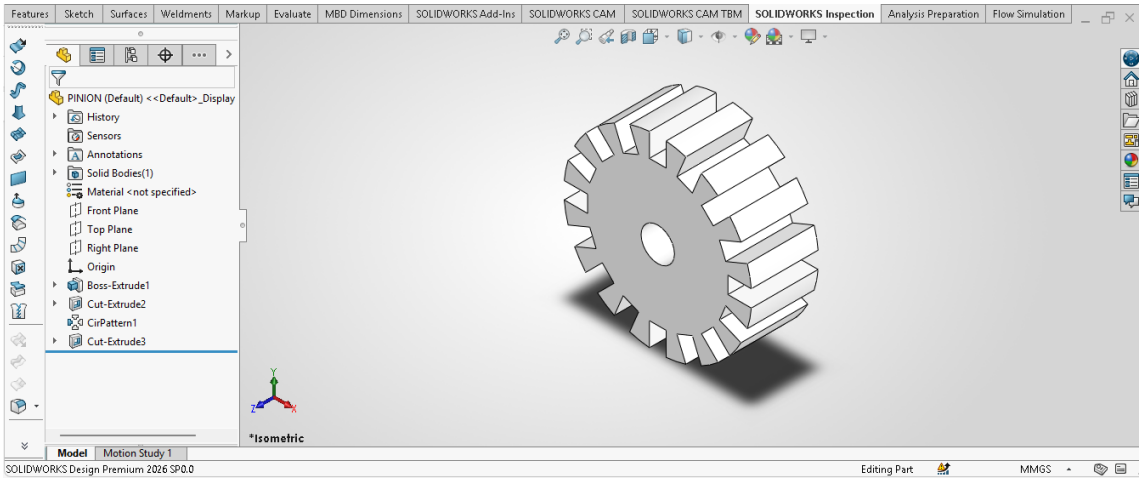


Figure 5: PINION and its feature tree – Boss-Extrude1, Cut-Extrude2, CirPattern1 (16 teeth), Cut-Extrude3 (bore). Material not yet assigned in the model.

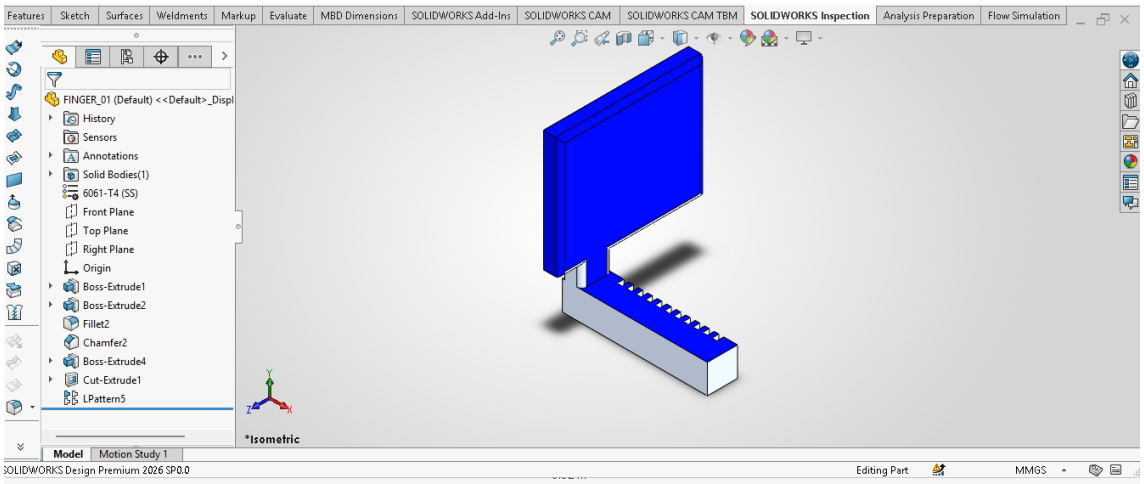


Figure 6: FINGER_01 and its feature tree – Boss-Extrudes, Fillet2, Chamfer2, Cut-Extrude1, LPattern5 (rack teeth). Material: 6061-T4 (SS) in the model.

5. Assembly (Question 4)

Mate	Purpose
Housing_02 fixed	Reference part
Pinion: Concentric + Coincident	Pinion sits on the channel floor on the 4 mm hole and can only spin
Each finger: Coincident to floor	Finger slides in the horizontal plane only
Each finger: Coincident to housing wall	Finger slides along the channel only (one degree of freedom)
Rack-Pinion mate × 2 (pitch diameter 20 mm)	Links each finger to the pinion; one mate is reversed so the fingers move toward each other

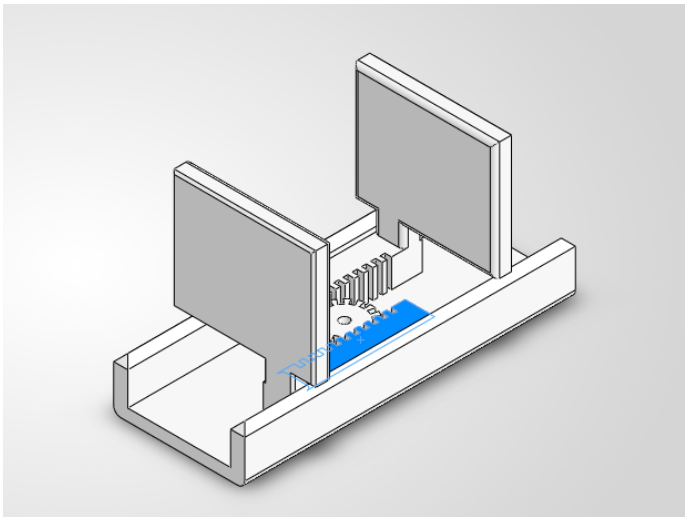


Figure 7: Assembled gripper (isometric).

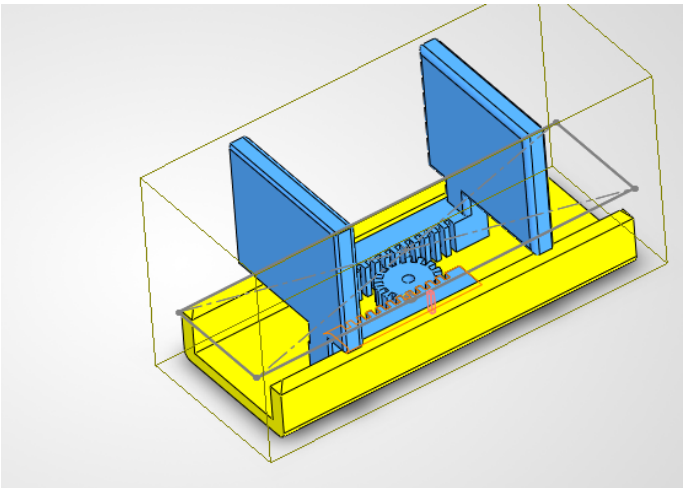


Figure 8: Assembly with fingers in the closed position, mate display on.

5.1 Interference check

Interference Detection in SolidWorks reported eight small overlaps between the pinion and the two fingers, with a total volume of about 115.8 mm^3 and a largest single item of 26.65 mm^3. They are small compared with the part sizes and are most likely caused by the simplified rectangular teeth. They must be re-checked after the second finger is corrected (Section 8).

ID	Volume (mm^3)	Components
Interference1	26.65	PINION-2 / FINGER_01-4
Interference2	25.81	PINION-2 / FINGER_01-4
Interference3	23.27	PINION-2 / FINGER_01-3
Interference4	23.03	PINION-2 / FINGER_01-3
Interference5	6.35	PINION-2 / FINGER_01-4
Interference6	6.23	PINION-2 / FINGER_01-3
Interference7	2.71	PINION-2 / FINGER_01-4
Interference8	1.76	PINION-2 / FINGER_01-3
Total	**115.81**	

6. Gripping Force Calculation (Question 4)

6.1 Assumptions

Friction coefficient between finger and payload μ = 0.3 (conservative). Safety factor SF = 1.5 (minimum required). g = 9.81 m/s^2. Two friction contacts, one per finger. The payload is held against gravity by friction only, with no acceleration.

6.2 Required force per finger

Friction holds the load: 2 × μ × F = SF × m × g, so **F = SF × m × g / (2 × μ)**.

Payload	Weight (m × g)	Force per finger
5 kg	49.05 N	1.5 × 49.05 / 0.6 = 122.6 N (about 123 N)
10 kg	98.10 N	1.5 × 98.10 / 0.6 = 245.3 N (about 245 N)

A design load of **250 N per finger** is used for the stress checks. If the robot accelerates the payload at about 1 g, the required force doubles; that case is outside this analysis.

6.3 Actuator torque at the pinion

Both racks load the pinion, so T = 2 × F × r with pitch radius r = 10 mm = 0.010 m.

Payload	Torque
10 kg	T = 2 × 245 × 0.010 ≈ 4.9 N·m
5 kg	T = 2 × 123 × 0.010 ≈ 2.5 N·m

A small N20 gear motor delivers well under 1 N·m, so it cannot drive this gripper at full payload. The actuation interface (the 4 mm pinion bore) must accept a higher-torque geared motor or servo. Rubber finger pads (μ ≈ 0.6) would halve the required force and torque.

7. Static Stress Analysis (Question 5)

7.1 Status

The ANSYS static structural study has not been run yet. The sections below give a hand check, the prepared FEA setup and a results table that will be filled in after the simulation.

7.2 Hand check (approximate, not a replacement for FEA)

Check	Method	Result	Allowable	Factor of safety
Finger plate	Cantilever: σ = 6·F·L / (b·t²), F = 250 N, L ≈ 40 mm (estimated), b = 50 mm, t = 8 mm	≈ 19 MPa	Al 6061-T4: ≈ 145 MPa yield Al 6061-T6: ≈ 275 MPa yield	≈ 7.7 (T4) ≈ 14 (T6)
Pinion tooth	Lewis: σ = Ft / (b·m·Y) = 250 / (8 × 1.25 × 0.295)	≈ 85 MPa	Al 6061-T4: ≈ 145 MPa Steel: higher	≈ 1.7 (T4) higher in steel

The lever arm L is an estimate, and the peak stress at the neck fillet of the finger is expected to be higher than this simple beam value. The pinion check shows an aluminium pinion would barely pass the 1.5 requirement and a PLA pinion would not pass, so the pinion should be made of steel.

7.3 Prepared ANSYS setup

Item	Planned value
Study type	Static Structural (ANSYS Workbench)
Part analysed	FINGER_01 (critical component); pinion as a second study
Material	Aluminium 6061 (T4 or T6, to be fixed and stated); steel for the pinion
Fixed support	End face of the rack bar
Load	250 N on the inner gripping face of the plate, normal to the face
Mesh	Refined at the neck fillet; mesh convergence check on peak stress
Results to report	Equivalent (von Mises) stress, total deformation, equivalent elastic strain, factor of safety

7.4 Results (to be completed)

Quantity	FEA result	Requirement
Max equivalent stress	Pending	Below yield / 1.5
Max total deformation	Pending	Small compared with the 60 mm plate
Max equivalent strain	Pending	Elastic range
Minimum factor of safety	Pending	At least 1.5

8. Known Issues and Corrective Actions

Issue in v1	Effect	Planned fix
Second finger was placed without the 180° rotation about the pinion axis	Rack mesh on that side is not set up correctly; the assembly motion and interference results may not be reliable	Rotate (or mirror) the second finger, re-apply the rack-pinion mate, and re-run Interference Detection
Rectangular teeth instead of involute	Small tooth-tip interference, approx. 115.8 mm^3 in total	Model involute tooth profiles for pinion and racks
No ANSYS simulation	No verified stress or deformation values	Run the static structural study from Section 7.3
Material not assigned to the pinion; finger set to 6061-T4	Yield assumption must match the model	Assign materials consistently (steel pinion, stated aluminium grade for fingers)
No STEP export	Model opens only in SolidWorks	Export the assembly as STEP

9. Conclusion

The rack-and-pinion mechanism gives symmetric, parallel opening and closing of two jaws from a single pinion. The model is parametric and uses extrude, cut, fillet, chamfer and pattern features with the required yellow and blue colouring.

For a 5–10 kg payload, with $\mu = 0.3$ and a safety factor of 1.5, each finger needs about 123–245 N of grip force, which requires about 2.5–4.9 N·m at the pinion. A small N20 motor is not sufficient, so a higher-torque geared motor or servo is needed. Hand checks suggest the finger plate has a comfortable margin and the pinion should be steel.

The design is not yet validated by simulation. The next version will correct the second-finger orientation, add involute teeth, and complete the ANSYS static analysis so that the stress, strain and deformation results can replace the hand estimates.

10. Future Work

Step	Goal
1. Fix assembly	Correct second-finger orientation and mates
2. Involute teeth	Remove tooth-tip interference
3. ANSYS static structural	Stress, strain, deformation and factor of safety for finger and pinion
4. Finger pads	Rubber pads ($\mu \approx 0.6$) to halve grip force
5. Actuator selection	Geared motor or servo for about 5 N·m